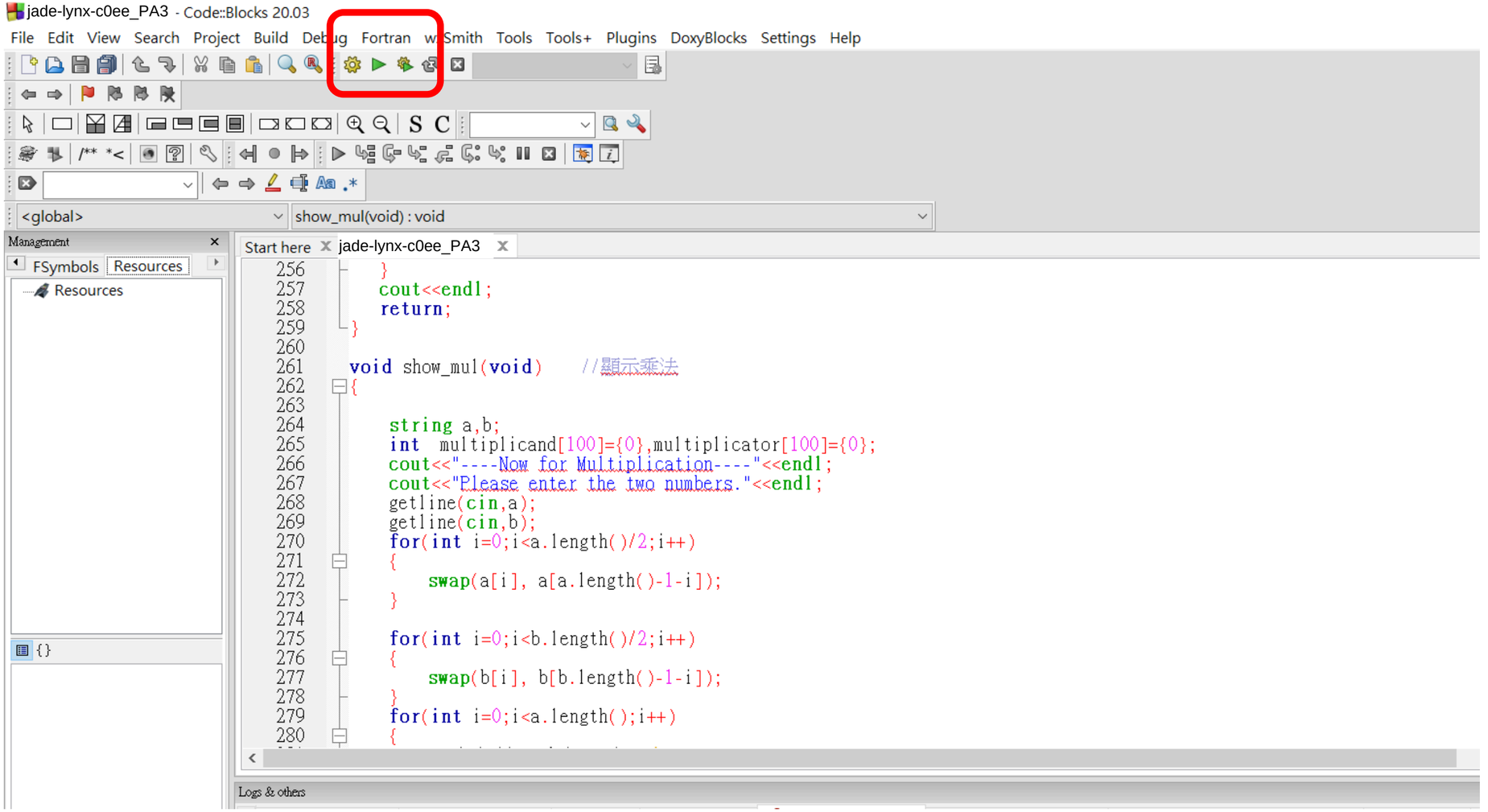
How to compile and execute your program



How to compile and execute your program

```
void show_prompt(void);

void show_add(void);
void add_calculator(int[],int[]);

void show_sub(void);
void sub_calculator(int[],int[]);

void show_mul(void);
void mul_calculator(int[],int[]);

void show_mean(int[],int[]);
```

顯示提示訊息

處理加法

處理減法

處理乘法

算平均

加法:

```
void add_calculator(int a[],int b[])  //加法
{
     int carry=0,c[100]={0},start;
    for(int i=0;i<100;i++)
    {
        c[i]=a[i]+b[i]+carry;
        carry=c[i]/10;
        c[i]%=10;
    }
    for(int i=99;i>=0;i--)
    {
        if(c[i]!=0)
        {
             start=i;
             break;
        }
        if(i==0)  start=0;
    }
    for(int i=start;i>=0;i--)
    {
        cout<<c[i];
    }
    cout<<endl;
    return;

}
```

將a陣列與b陣列一格對一格相加再加上carry存到c陣列

最後印出結果

減法:

```
void sub_calculator(int a[],int b[])   //減法
{
    int borrow=0,c[100]={0},start;

   for(int i=0;i<100;i++)
   {
       c[i]=a[i]-b[i]-borrow;
       if(c[i]<0)
       {
            borrow=1;
            c[i]+=10;
       }
       else
       {
           borrow=0;
       }

   }
   for(int i=99;i>=0;i--)
   {
       if(c[i]!=0)
       {
            start=i;
            break;
       }
       if(i==0)  start=0;
   }
   for(int i=start;i>=0;i--)
   {
       cout<<c[i];
   }
   cout<<endl;
   return;
}
```

將a陣列與b陣列一格對一格相減再減去borrow存到c陣列，如果c[i]小於0，代借位發生，borrow要是1，然後c[i]在加10，如果沒發生借位，borrow則是0

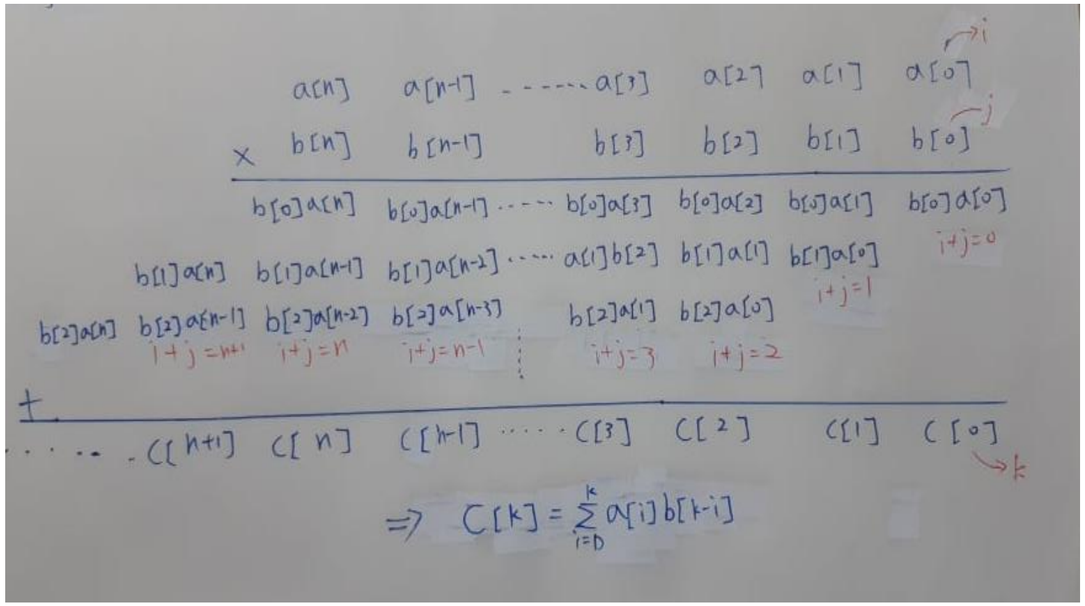
最後印出結果

The hardness of this assignment and how you overcome it

我覺得大數的乘法是最難的，拆成一位一位後根本不知道要著麼做，但後來模擬一般的乘法運算後找到了這個規律

```
void mul_calculator(int a[],int b[])   //乘法
{
    int carry=0,c[200]={0},start;
    for(int i=0;i<100;i++)  //乘法規律
        for(int j=0;j<100;j++)
    {
        c[i+j]=a[i]*b[j]+c[i+j];
    }
    for(int i=0;i<100;i++)
    {
        c[i]+=carry;
        carry=c[i]/10;
        c[i]%=10;
    }
    for(int i=99;i>=0;i--)
    {
        if(c[i]!=0)
        {
            start=i;
            break;
        }
        if(i==0)  start=0;
    }
    for(int i=start;i>=0;i--)
    {
        cout<<c[i];
    }
    cout<<endl;
    return;
}
```

之後再一次進位



The completion of the assignment: all

jade-lynx-c0ee_PA3

```
----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
1
----Now for Addition----
Please enter the two numbers.
11112222333344445555
77776666555544443333
Result: 88888888888888888888
Mean: 4444444444444444444

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
5
Error! Please try again.

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
3
----Now for Multiplication----
Please enter the two numbers.
1111111111
5
Result: 5555555555
Mean: 555555558

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
0
Good bye!
```

jade-lynx-c0ee_PA3

```
----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
2
----Now for Subtraction ----
Please enter the two numbers.
111111111
33333333
Result: 77777778
Mean: 72222222

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
5
Error! Please try again.

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
3
----Now for Multiplication----
Please enter the two numbers.
29292929
300000003
Result: 8787878787878787
Mean: 164646466

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
0
Good bye!
```

jade-lynx-c0ee_PA3 .exe"

```
----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
2
----Now for Subtraction ----
Please enter the two numbers.
9999999999
1212121212
Result: 8787878787
Mean: 5606060605

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
4
Error! Please try again.

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
3
----Now for Multiplication----
Please enter the two numbers.
29292929
3
Result: 87878787
Mean: 14646466

----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
0
Good bye!
```